

Formulário

Sejam $f, g: I \rightarrow \mathbb{R}$ funções deriváveis no intervalo real I .

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

$$(f(x) \times g(x))' = f'(x)g(x) + g'(x)f(x)$$

$$(f[g(x)])' = g'(x)f'[g(x)]$$

$$\left[\frac{f(x)}{g(x)} \right]' = \frac{f'(x)g(x) - g'(x)f(x)}{g^2(x)}$$

$$[f^{-1}(y)]' = \frac{1}{f'[f^{-1}(y)]}$$

Sejam $f: I \rightarrow \mathbb{R}$ uma função derivável no intervalo real I e $\mathcal{C}$ uma constante real arbitrária e $\{k, n\} \in \mathbb{R}$.

$$1.P \quad k = kx + \mathcal{C}$$

$$3.P \quad \frac{f'(x)}{f(x)} = \ln |f(x)| + \mathcal{C}$$

$$5.P \quad a^{f(x)} f'(x) = \frac{a^{f(x)}}{\ln a} + \mathcal{C} \quad (a \in \mathbb{R}^+ \setminus \{1\})$$

$$7.P \quad f'(x) \sin(f(x)) = -\cos(f(x)) + \mathcal{C}$$

$$9.P \quad f'(x) \operatorname{cosec}^2(f(x)) = -\cotg(f(x)) + \mathcal{C}$$

$$11.P \quad f'(x) \operatorname{cosec}(f(x)) = \ln |\operatorname{cosec}(f(x)) - \cotg(f(x))| + \mathcal{C}$$

$$13.P \quad \frac{-f'(x)}{\sqrt{1 - f^2(x)}} = \arccos(f(x)) + \mathcal{C}$$

$$15.P \quad \frac{-f'(x)}{1 + f^2(x)} = \operatorname{arccotg}(f(x)) + \mathcal{C}$$

$$17.P \quad f'(x) \operatorname{ch}(f(x)) = \operatorname{sh}(f(x)) + \mathcal{C}$$

$$19.P \quad \frac{f'(x)}{\operatorname{ch}^2(f(x))} = \operatorname{th}(f(x)) + \mathcal{C}$$

$$21.P \quad \frac{f'(x)}{\sqrt{f^2(x) + 1}} = \operatorname{argsh}(f(x)) + \mathcal{C}$$

$$23.P \quad \frac{f'(x)}{1 - f^2(x)} = \operatorname{argth}(f(x)) + \mathcal{C}$$

$$2.P \quad f'(x)f^n(x) = \frac{f^{n+1}(x)}{n+1} + \mathcal{C} \quad (n \neq -1)$$

$$4.P \quad f'(x)e^{f(x)} = e^{f(x)} + \mathcal{C}$$

$$6.P \quad f'(x) \cos(f(x)) = \sin(f(x)) + \mathcal{C}$$

$$8.P \quad f'(x) \sec^2(f(x)) = \operatorname{tg}(f(x)) + \mathcal{C}$$

$$10.P \quad f'(x) \sec(f(x)) = \ln |\sec(f(x)) + \operatorname{tg}(f(x))| + \mathcal{C}$$

$$12.P \quad \frac{f'(x)}{\sqrt{1 - f^2(x)}} = \operatorname{arcsen}(f(x)) + \mathcal{C}$$

$$14.P \quad \frac{f'(x)}{1 + f^2(x)} = \operatorname{arctg}(f(x)) + \mathcal{C}$$

$$16.P \quad \frac{f'(x)}{|f(x)| \sqrt{f^2(x) - 1}} = \operatorname{arcsec}(f(x)) + \mathcal{C}$$

$$18.P \quad f'(x) \operatorname{sh}(f(x)) = \operatorname{ch}(f(x)) + \mathcal{C}$$

$$20.P \quad \frac{f'(x)}{\operatorname{sh}^2(f(x))} = -\operatorname{coth}(f(x)) + \mathcal{C}$$

$$22.P \quad \frac{f'(x)}{\sqrt{f^2(x) - 1}} = \operatorname{argch}(f(x)) + \mathcal{C}$$

$$24.P \quad \frac{f'(x)}{1 - f^2(x)} = \operatorname{argcoth}(f(x)) + \mathcal{C}$$

$$\sin \frac{\pi}{6} = \cos \frac{\pi}{3} = \frac{1}{2}$$

$$\sin \frac{\pi}{3} = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

$$\cos^2 \alpha + \sin^2 \alpha = 1$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}$$

$$\sec \alpha = \frac{1}{\cos \alpha}$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \sin \beta \cos \alpha$$

$$\sin^2 \alpha = \frac{1 - \cos 2\alpha}{2}$$

$$\operatorname{cosec} \alpha = \frac{1}{\sin \alpha}$$

$$\sec^2 \alpha = 1 + \operatorname{tg}^2 \alpha$$

$$\operatorname{cosec}^2 \alpha = 1 + \cotg^2 \alpha$$

$$\operatorname{ch} u = \frac{e^u + e^{-u}}{2}$$

$$\operatorname{sh} u = \frac{e^u - e^{-u}}{2}$$

$$\operatorname{ch}^2 u - \operatorname{sh}^2 u = 1$$

$$\operatorname{th} u = \frac{\operatorname{sh} u}{\operatorname{ch} u}$$

$$\operatorname{ch}(u \pm v) = \operatorname{ch} u \operatorname{ch} v \pm \operatorname{sh} u \operatorname{sh} v$$

$$\operatorname{ch}^2 u = \frac{\operatorname{ch}(2u) + 1}{2}$$

$$\operatorname{coth} u = \frac{1}{\operatorname{th} u}$$

$$\operatorname{sh}(u \pm v) = \operatorname{sh} u \operatorname{ch} v \pm \operatorname{sh} v \operatorname{ch} u$$

$$\operatorname{sh}^2 u = \frac{\operatorname{ch}(2u) - 1}{2}$$

Primitivação por Substituição

Na lista de substituições que se segue, a , b e c são constantes reais arbitrárias. A notação $R(\dots)$ indica uma função racional dos monómios que se encontram dentro dos parêntesis. Na coluna da esquerda, figuram diferentes tipos de funções primitiváveis. Na coluna da direita sugere-se, em cada caso, uma substituição adequada à função indicada na coluna da esquerda.

Tipo de Função

Substituição

1. $\frac{1}{(x^2 + a^2)^k}$, $k \in \mathbb{N}$, $k > 1$

$$x = a \operatorname{tg} t$$

2. $R(a^{rx}, a^{sx}, \dots)$

$$a^{mx} = t \text{ com } m = \text{m.d.c.}(r, s, \dots)$$

3. $R(\log_a x)$

$$t = \log_a x$$

4. $R\left(x, \left(\frac{ax+b}{cx+d}\right)^{p/q}, \left(\frac{ax+b}{cx+d}\right)^{r/s}, \dots\right)$

$$\frac{ax+b}{cx+d} = t^m \text{ com } m = \text{m.m.c.}(q, s, \dots)$$

5. $R\left(x, (ax+b)^{p/q}, (ax+b)^{r/s}, \dots\right)$

$$(ax+b) = t^m \text{ com } m = \text{m.m.c.}(q, s, \dots)$$

6. $R\left(x, x^{p/q}, x^{r/s}, \dots\right)$

$$x = t^m \text{ com } m = \text{m.m.c.}(q, s, \dots)$$

7. $R\left(x, \sqrt{a^2 - b^2 x^2}\right)$

$$x = \frac{a}{b} \operatorname{sen} t \text{ ou } x = \frac{a}{b} \cos t \text{ ou } x = \frac{a}{b} \operatorname{th} t$$

8. $R\left(x, \sqrt{a^2 + b^2 x^2}\right)$

$$x = \frac{a}{b} \operatorname{tg} t \text{ ou } x = \frac{a}{b} \operatorname{sh} t$$

9. $R\left(x, \sqrt{b^2 x^2 - a^2}\right)$

$$x = \frac{a}{b} \sec t \text{ ou } x = \frac{a}{b} \operatorname{ch} t$$

10. $R(\operatorname{sen} x, \cos x)$ com

(a) R ímpar em $\operatorname{sen} x$ isto é

$$R(-\operatorname{sen} x, \cos x) = -R(\operatorname{sen} x, \cos x)$$

$$\cos x = t$$

(b) R ímpar em $\cos x$ isto é

$$R(\operatorname{sen} x, -\cos x) = -R(\operatorname{sen} x, \cos x)$$

$$\operatorname{sen} x = t$$

(c) R par em $(\operatorname{sen} x, \cos x)$ isto é

$$R(-\operatorname{sen} x, -\cos x) = R(\operatorname{sen} x, \cos x)$$

$$\operatorname{tg} x = t, \text{ sendo então (supondo } x \in]0, \pi/2[)$$

$$\operatorname{sen} x = \frac{t}{\sqrt{1+t^2}}, \cos x = \frac{1}{\sqrt{1+t^2}}$$

(d) nos restantes casos (e até nos anteriores)

$$\operatorname{tg} \frac{x}{2} = t, \text{ sendo } \operatorname{sen} x = \frac{2t}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}$$

11. $R(\operatorname{sh} x, \operatorname{ch} x)$ com

(a) R ímpar em $\operatorname{sh} x$

$$\operatorname{ch} x = t$$

(b) R ímpar em $\operatorname{ch} x$

$$\operatorname{sh} x = t$$

(c) R par em $(\operatorname{sh} x, \operatorname{ch} x)$

$$\operatorname{th} x = t, \text{ sendo } \operatorname{sh} x = \frac{t}{\sqrt{1-t^2}}, \operatorname{ch} x = \frac{1}{\sqrt{1-t^2}}$$

(d) nos restantes casos (e até nos anteriores)

$$\operatorname{th} \frac{x}{2} = t, \text{ sendo } \operatorname{sh} x = \frac{2t}{1-t^2}, \operatorname{ch} x = \frac{1+t^2}{1-t^2}$$